

Numerical Analysis — Report

Assignment 1: Root-Finding Methods

1. Choice of Initial Guesses and Bracketing Intervals

- **Bisection Method:** For $f(x) = x - \cos(x)$ on $[0, \pi]$, $f(0) = -1 < 0$ and $f(\pi) \approx 4.14 > 0$. By the Intermediate Value Theorem, a root is guaranteed. The theoretical bound is $n \geq \lceil \ln(\pi/10^{-4})/\ln(2) \rceil = 15$ iterations. The code terminated at exactly 15 iterations.
- **Newton's & Modified Newton's:** For $f(x) = e^x - x - 1$, $x = 0$ is a double root ($m = 2$) where $f'(0) = 0$. An initial guess of $p_0 = 1.0$ ensures proximity to the root while avoiding division by zero. For $P(x) = x^4 - 10x^3 + 35x^2 - 50x + 24$, $p_0 = 0.5$ converges to root $p = 1.0$.
- **Fixed-Point Iteration:** For $x = \cos(x)$, $|g'(x)| = |-\sin(x)| \leq \sin(1) \approx 0.8415 < 1$ on $[0, 1]$. By the Contraction Mapping Theorem, this guarantees unique convergence.

2. Order of Convergence and Asymptotic Constants

Method	Problem / Root	Iters	CPU Time (s)	Empirical α	Theoretical α
Bisection	$x - \cos(x) = 0$	15	4.73e-05	1.00	1
Standard Newton (m=1)	$e^x - x - 1 = 0$	27	1.48e-04	$\approx 1.29 \rightarrow 1.00$	1
Modified Newton (m=2)	$e^x - x - 1 = 0$	5	3.60e-05	2.00	2
Fixed-Point	$x = \cos(x)$	27	3.96e-05	1.0000	1

- **Multiple Roots Degradation:** When $m > 1$, $f'(p) = 0$ causes standard Newton's error ratio to tend toward $(m-1)/m = 0.5$, degrading convergence to linear (27 iterations).
- **Modified Newton Restoration:** Scaling the step by $m = 2$ cancels first-order error terms, restoring quadratic convergence ($\alpha = 2.00$) in only 5 iterations (a 4.1x speedup).
- **Asymptotic Constant:** Fixed point yielded $\lambda \approx 0.67381$, matching the theoretical derivative $|g'(p)| = |-\sin(0.739085)| \approx 0.6736$.

3. Horner's Method vs. Direct Evaluation

Both schemes converged in 6 iterations since Horner's method preserves the exact mathematical sequence. While Python loop overhead is noticeable for small degree $n = 4$, Horner reduces total operations from $O(n^2)$ to $O(n)$, avoiding severe round-off error in high-degree polynomials.

4. Choice of Iteration Function in Fixed-Point Methods

Choosing $g_1(x) = \cos(x)$ yields $|g_1'(0.739)| \approx 0.6736 < 1$, which contracts errors and converges. Conversely, $g_2(x) = \arccos(x)$ gives $|g_2'(0.739)| \approx 1.484 > 1$, which magnifies errors and diverges.